Clock test on LATENT residuals — TL;DR

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Periods: T=2 | T=5 | T=10 | T=50 | T=100 | T=1000

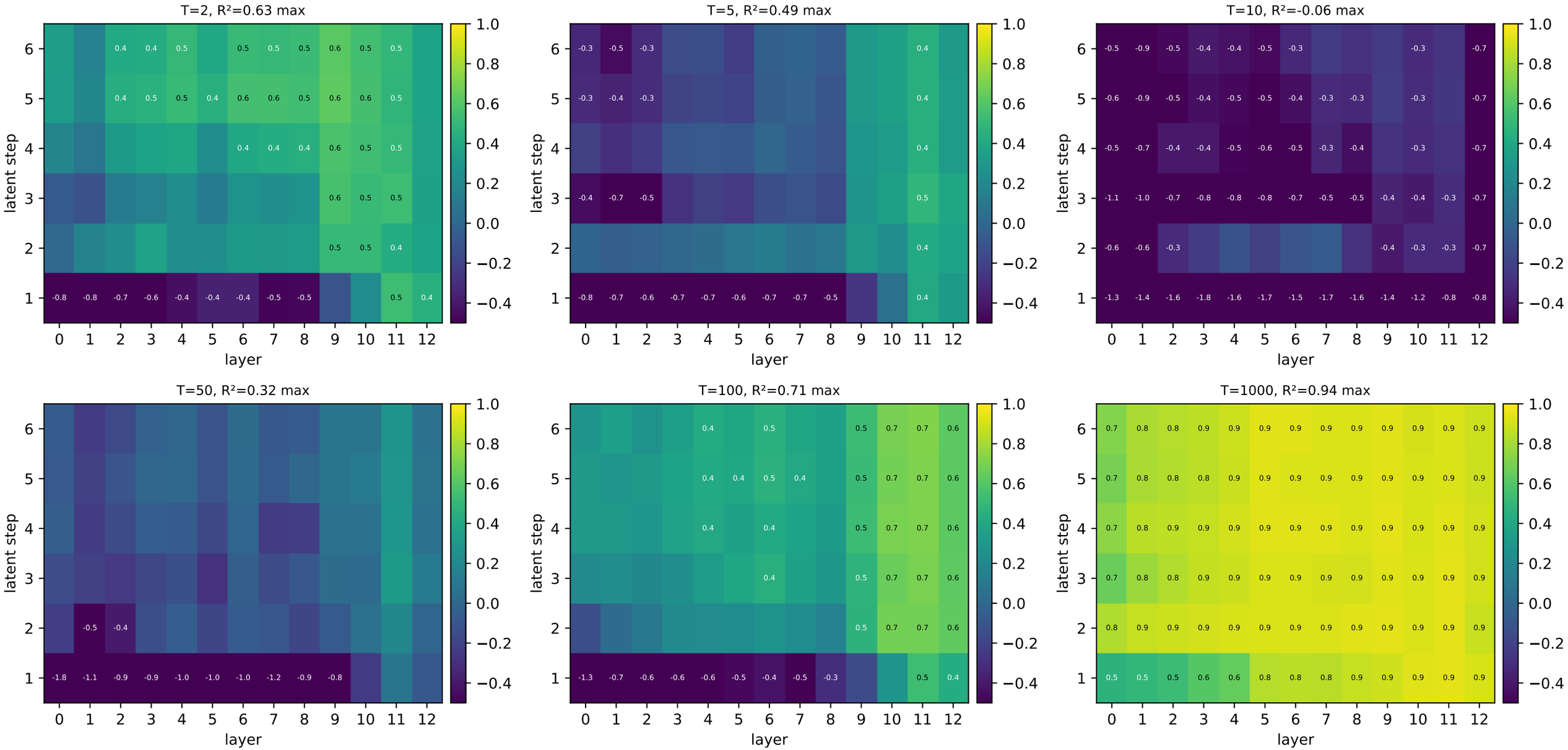
=== vary_numerals (Subtraction) ===
 T      best(step,L)  R²(a)  max(a)/T  Sub|resid|  Add|resid|
      2  (5, 9)       +0.63   99.00      0.746      0.746
      5  (3,11)       +0.49   39.60      1.089      1.652
     10  (2, 7)       -0.06   19.80      1.447      1.512
     50  (3,11)       +0.32   3.96      1.319      1.718
    100  (6,11)       +0.71   1.98      0.592      1.692
   1000  (4, 9)       +0.94   0.20      0.011      0.651
  ⇒ best non-trivial closure: T=100, |resid|=0.592 rad  (MODERATE)

=== vary_both_2digit (Subtraction) ===
 T      best(step,L)  R²(a)  max(a)/T  Sub|resid|  Add|resid|
      2  (3,11)       +0.74   49.50      0.550      0.550
      5  (5, 9)       +0.74   19.80      0.680      1.637
     10  (3,11)       +0.21   9.90      1.290      1.475
     50  (4,11)       +0.77   1.98      0.450      1.850
    100  (3,11)       +0.92   0.99      0.267      1.964
   1000  (4, 9)       +0.96   0.10      0.001      0.392
  ⇒ best non-trivial closure: T=100, |resid|=0.267 rad  (STRONG)

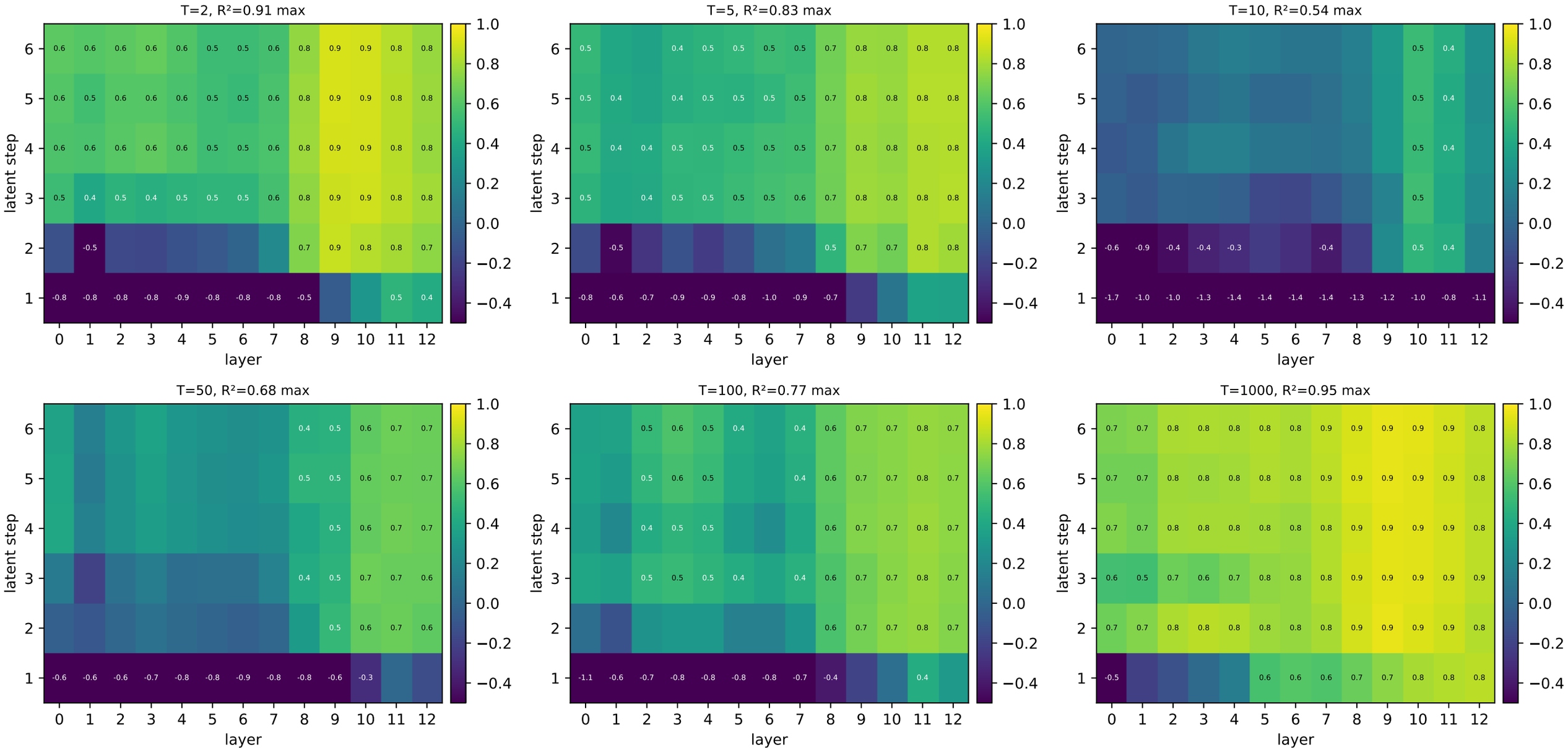
=== vary_a_2digit (Subtraction) ===
 T      best(step,L)  R²(a)  max(a)/T  Sub|resid|  Add|resid|
      2  (4,11)       +0.97   49.50      0.000      0.000
      5  (1,12)       +0.93   19.80      0.000      0.000
     10  (6,11)       +0.74   9.90      0.000      0.000
     50  (6,11)       +0.95   1.98      0.000      2.513
    100  (6,11)       +0.97   0.99      0.000      1.257
   1000  (6,10)       +0.98   0.10      0.000      0.126
  ⇒ best non-trivial closure: T=10, |resid|=0.000 rad  (STRONG)

=== vary_b_2digit (Subtraction) ===
 T      best(step,L)  R²(a)  max(a)/T  Sub|resid|  Add|resid|
      2  (1, 0)       +1.00   49.50      0.000      0.000
      5  (1, 0)       -37.22   19.80      0.000      1.653
     10  (1, 0)       -0.51   9.90      0.000      1.735
     50  (1, 0)       -4.57   1.98      0.000      1.735
    100  (1, 0)       -9.47   0.99      0.000      1.758
   1000  (1, 0)       -14.47   0.10      0.000      0.657
  ⇒ best non-trivial closure: T=2, |resid|=0.000 rad  (STRONG)
```

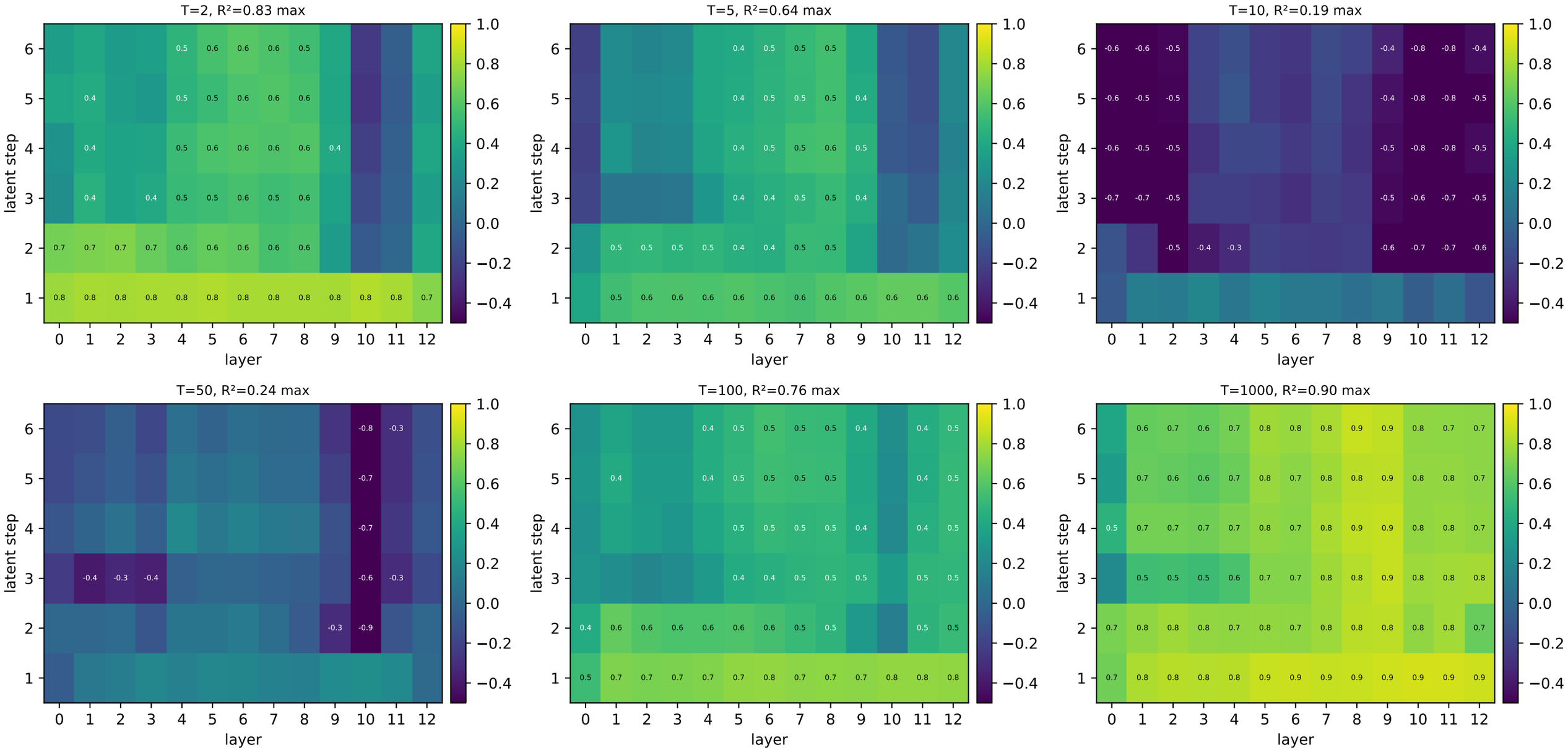
vary_numerals — helix R² for operand a (latent residual)



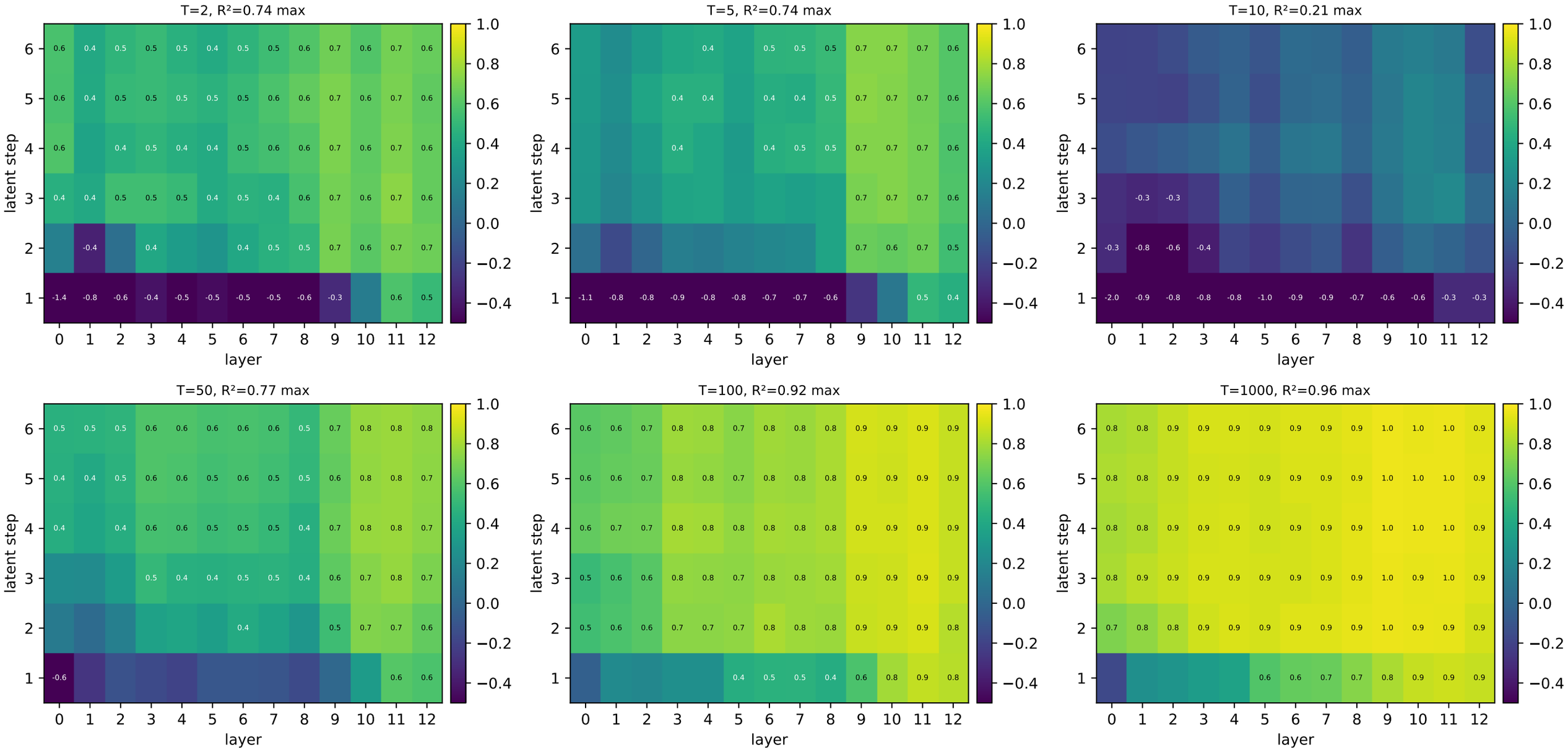
vary_numerals — helix R² for operand b (latent residual)



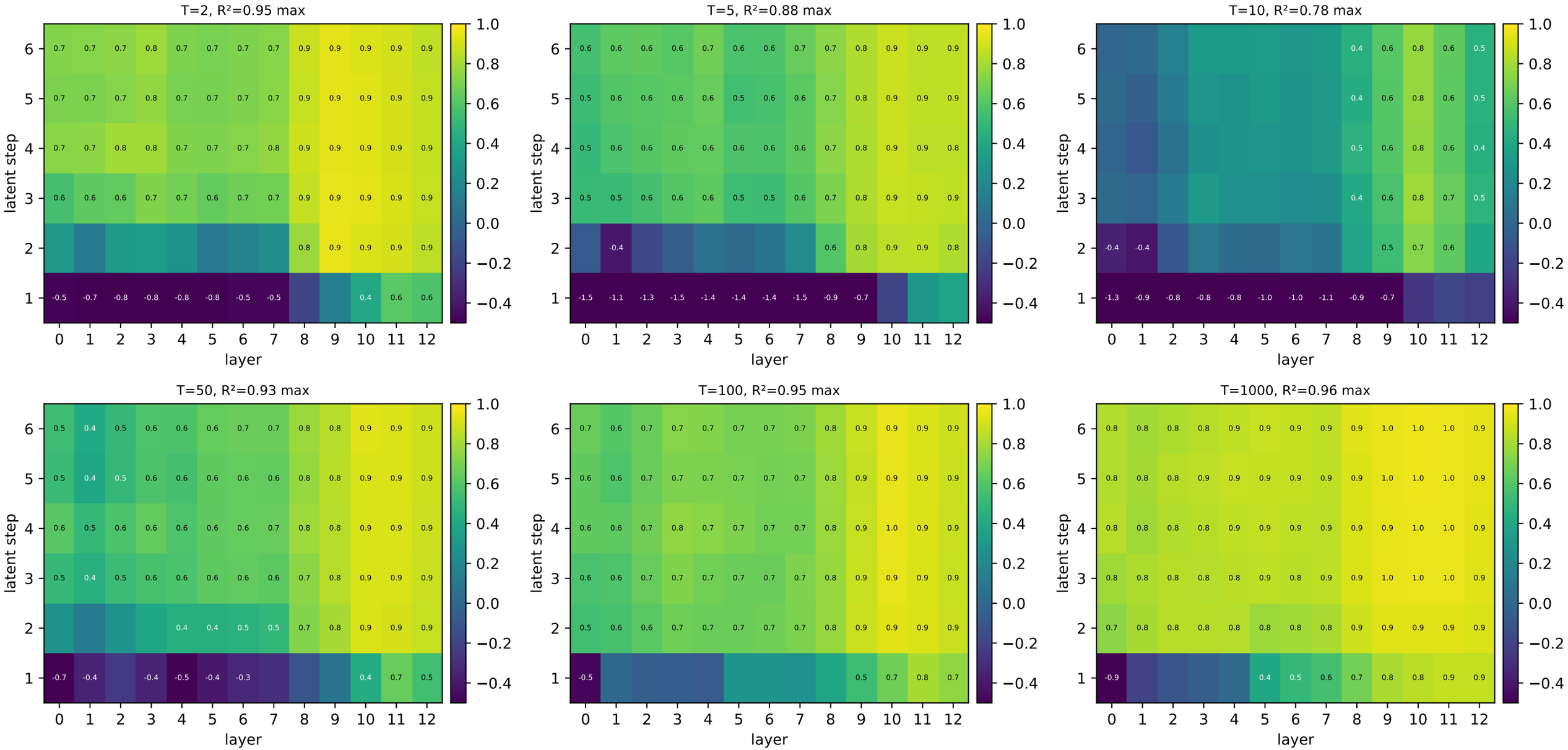
vary_numerals — helix R² for operand gold (latent residual)



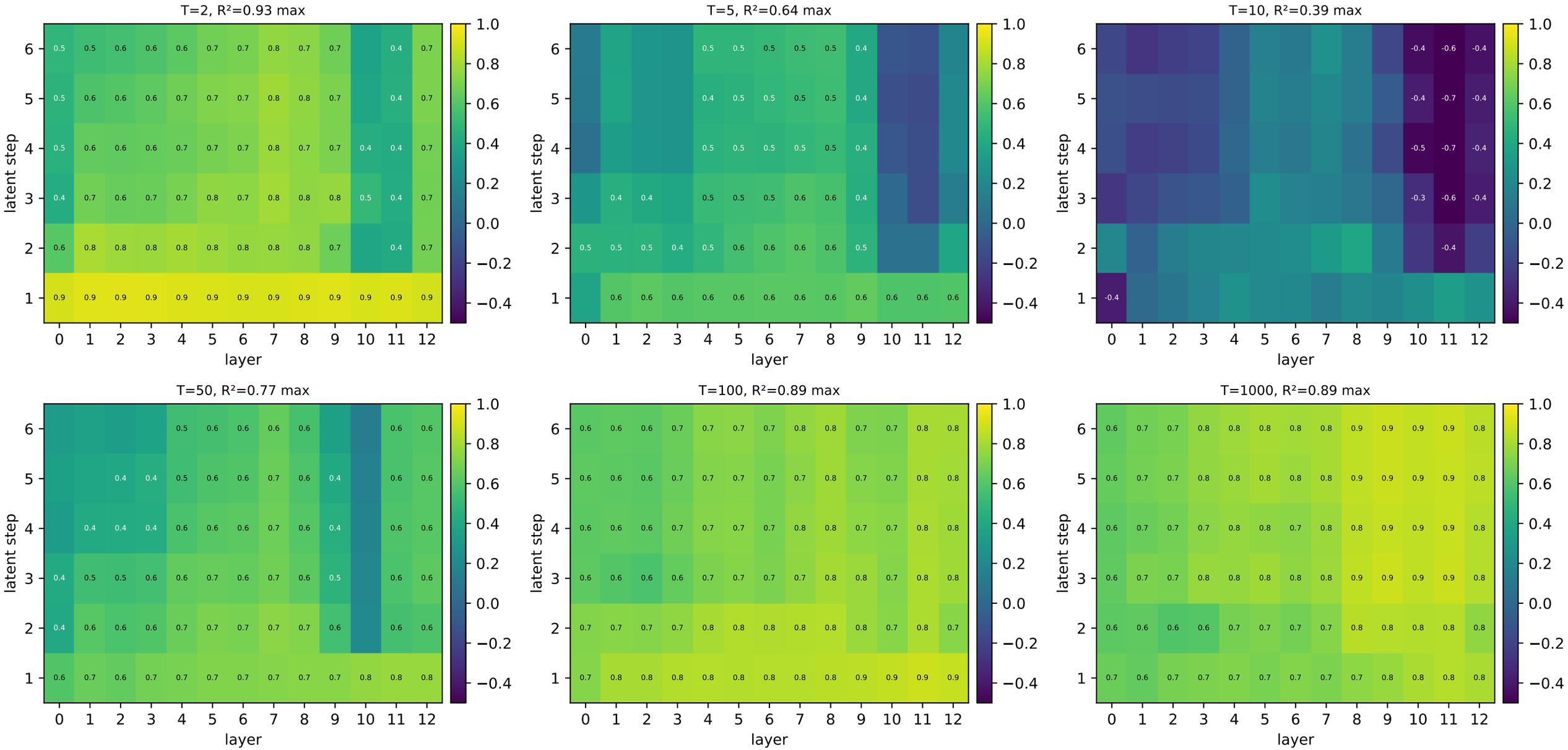
vary_both_2digit — helix R² for operand a (latent residual)



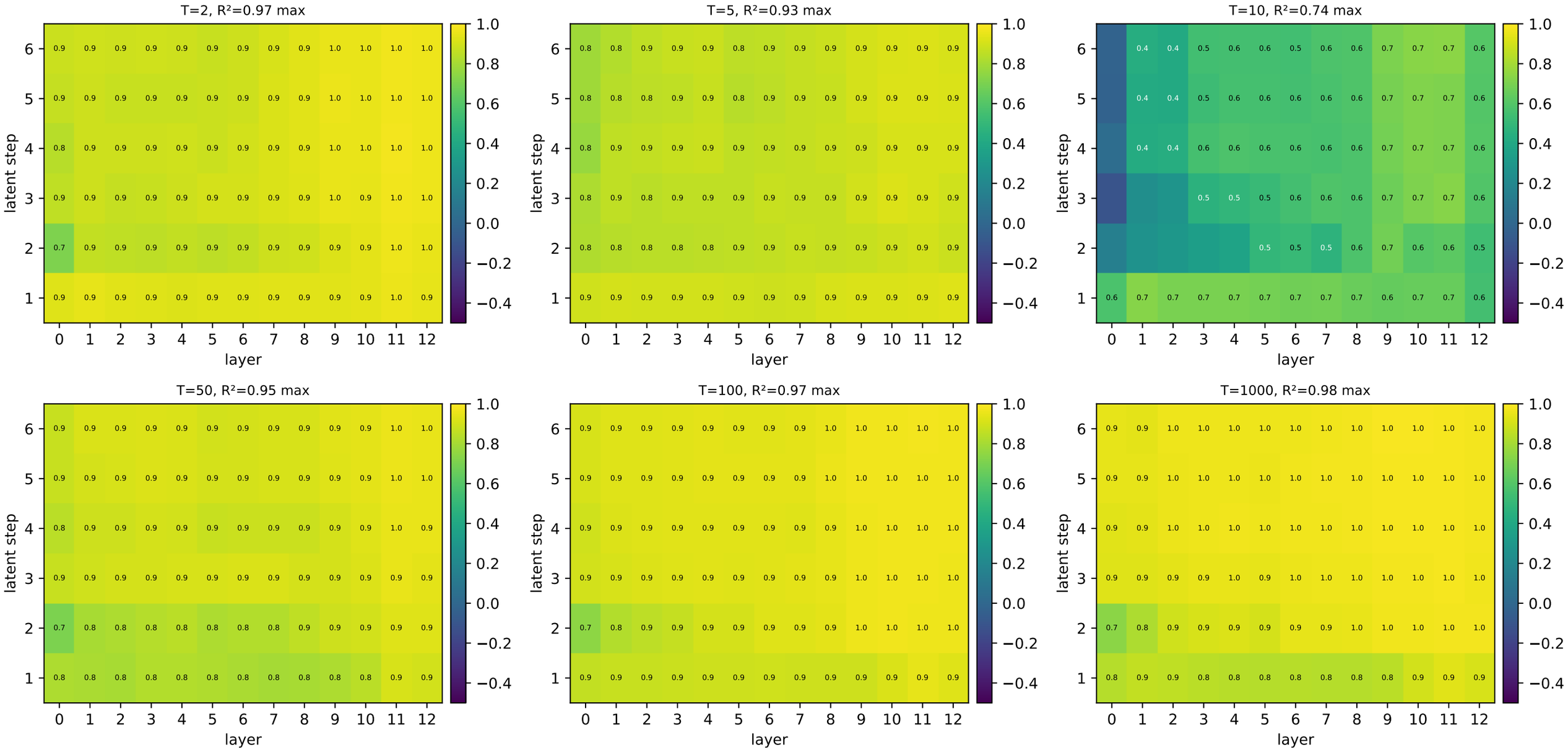
vary_both_2digit — helix R² for operand b (latent residual)



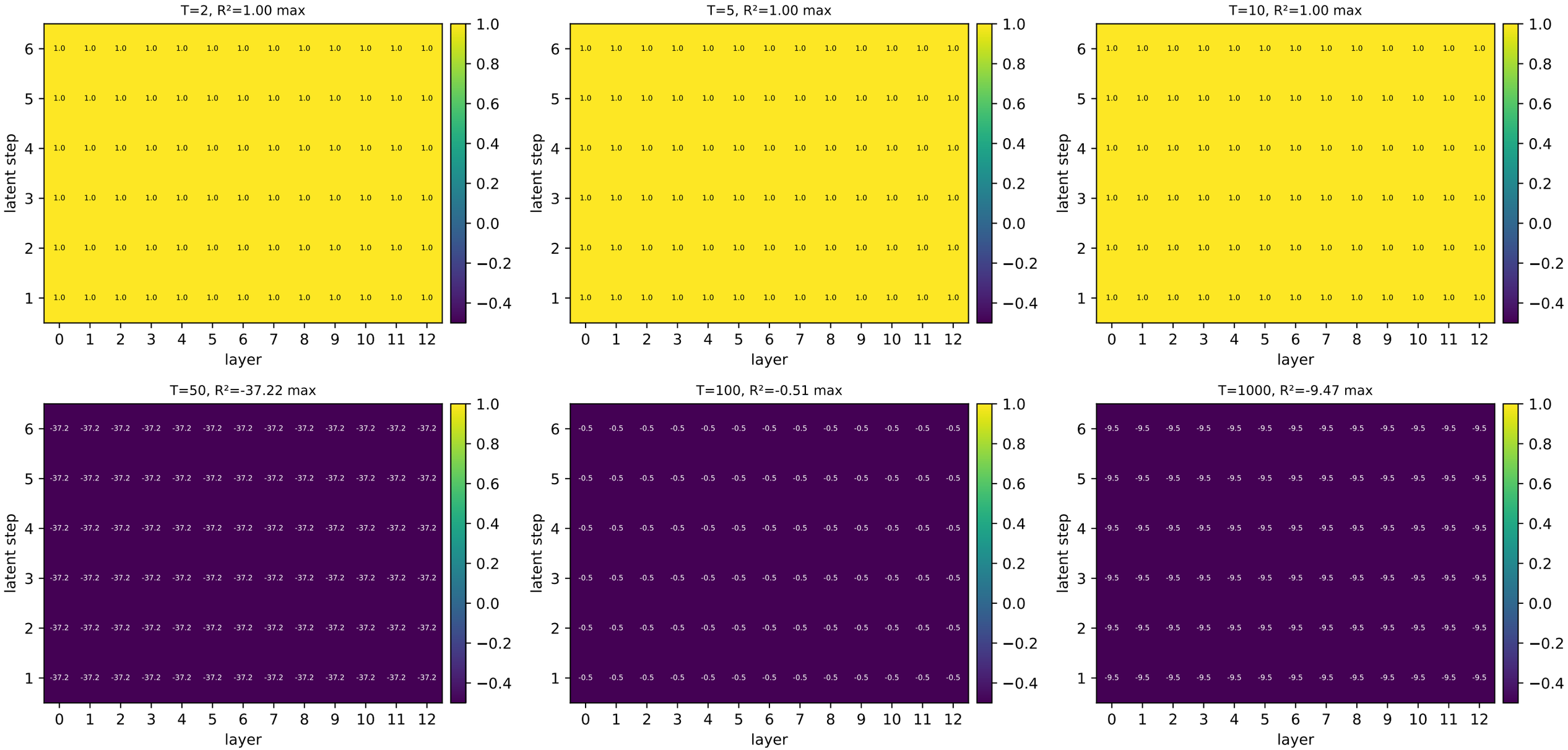
vary_both_2digit — helix R² for operand gold (latent residual)



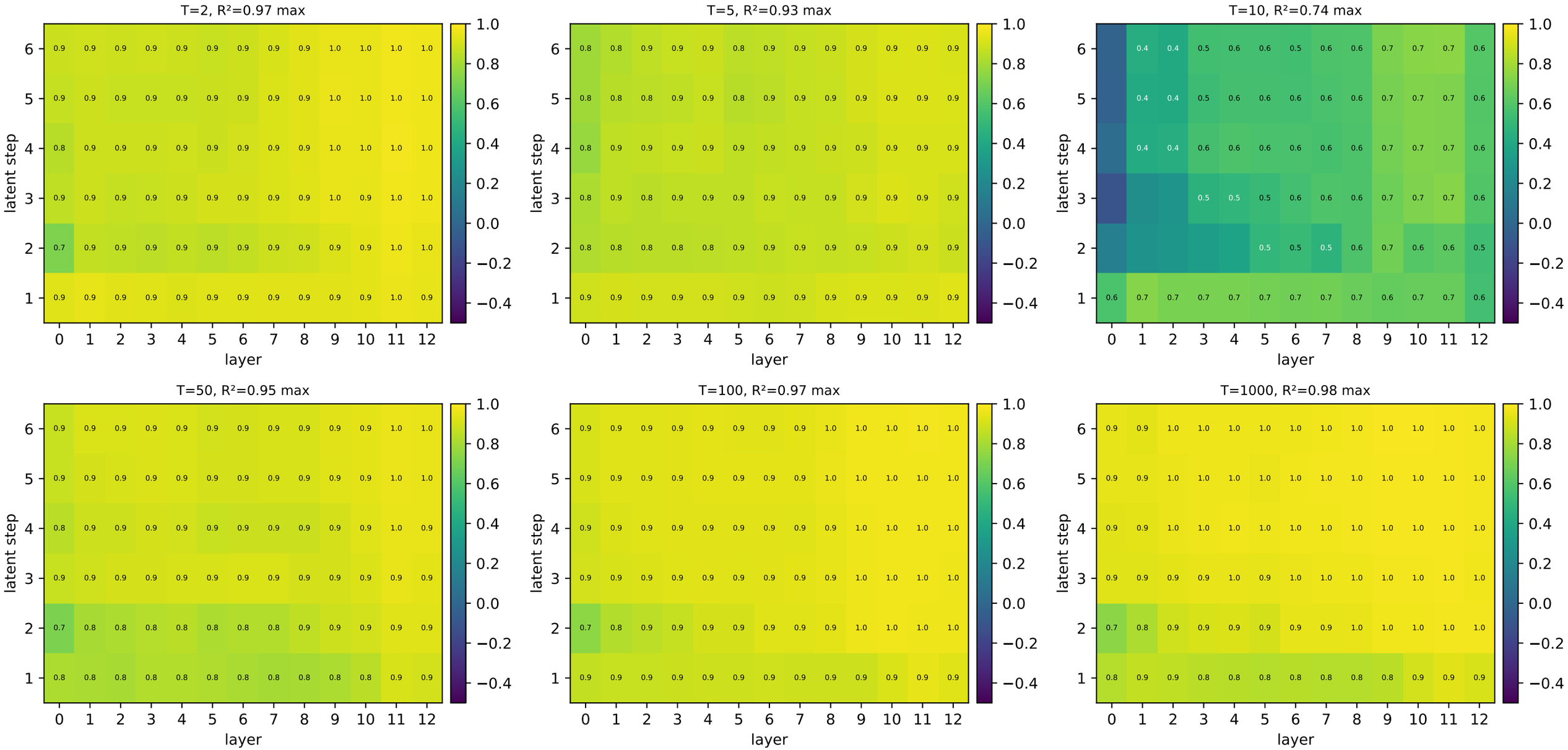
vary_a_2digit — helix R² for operand a (latent residual)



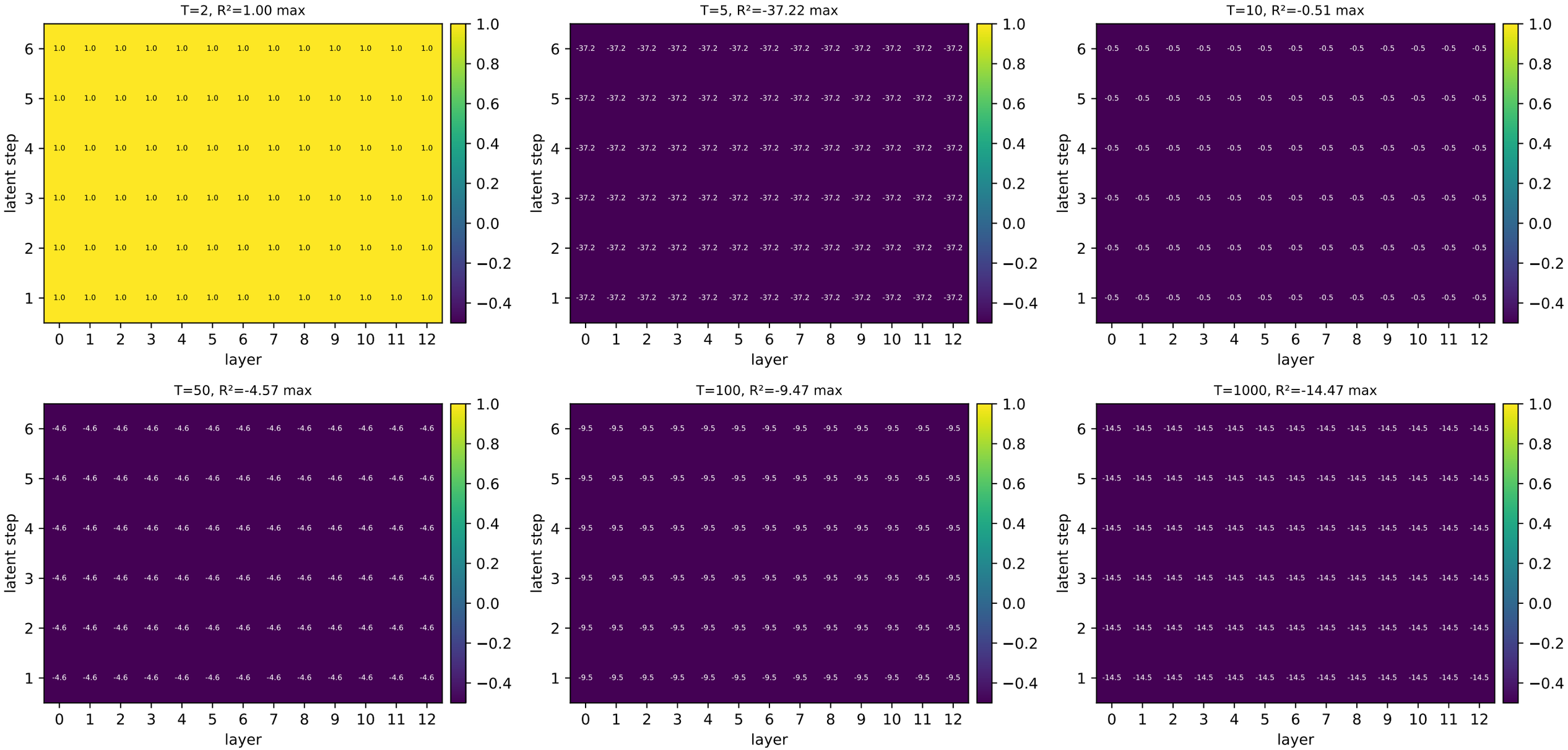
vary_a_2digit — helix R² for operand b (latent residual)



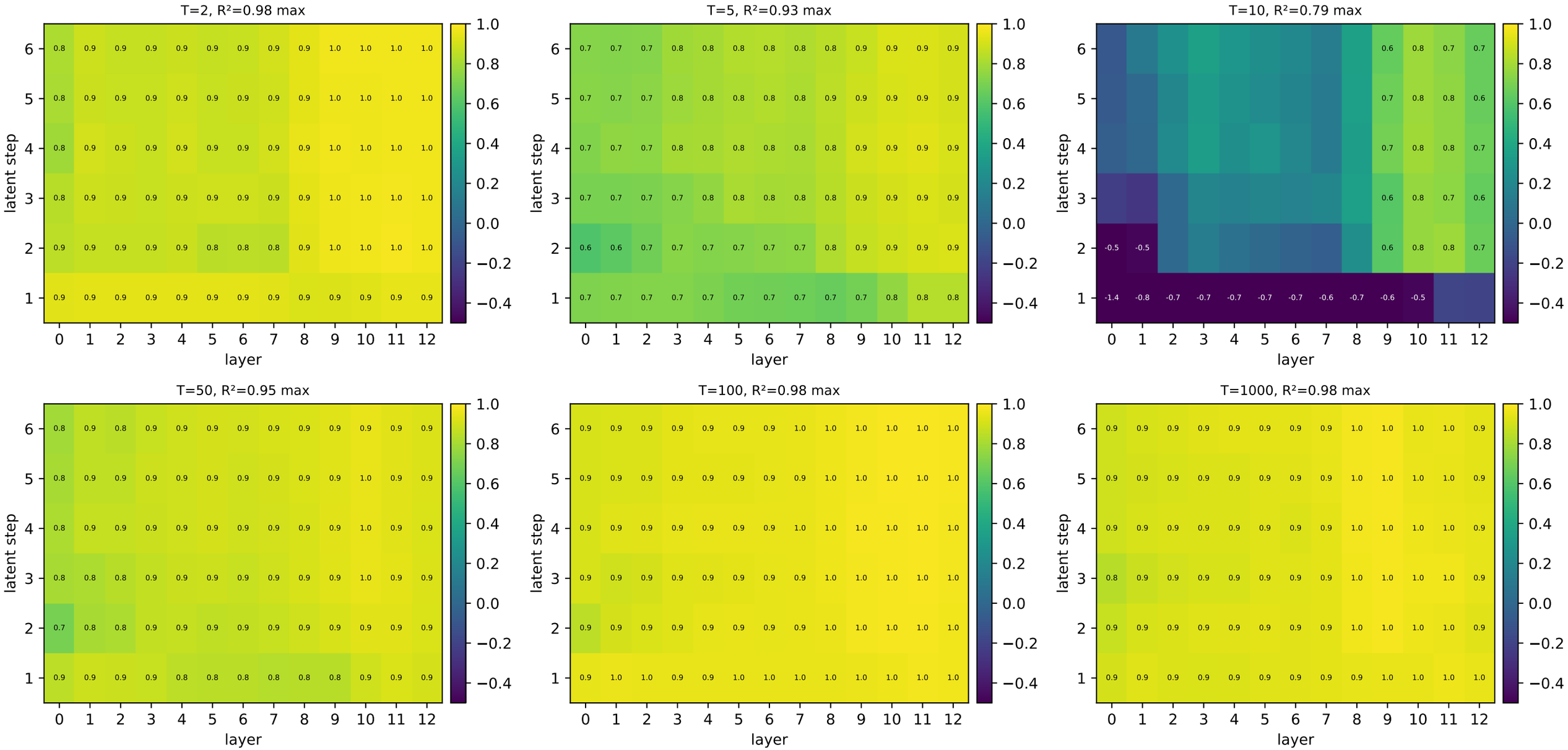
vary_a_2digit — helix R² for operand gold (latent residual)



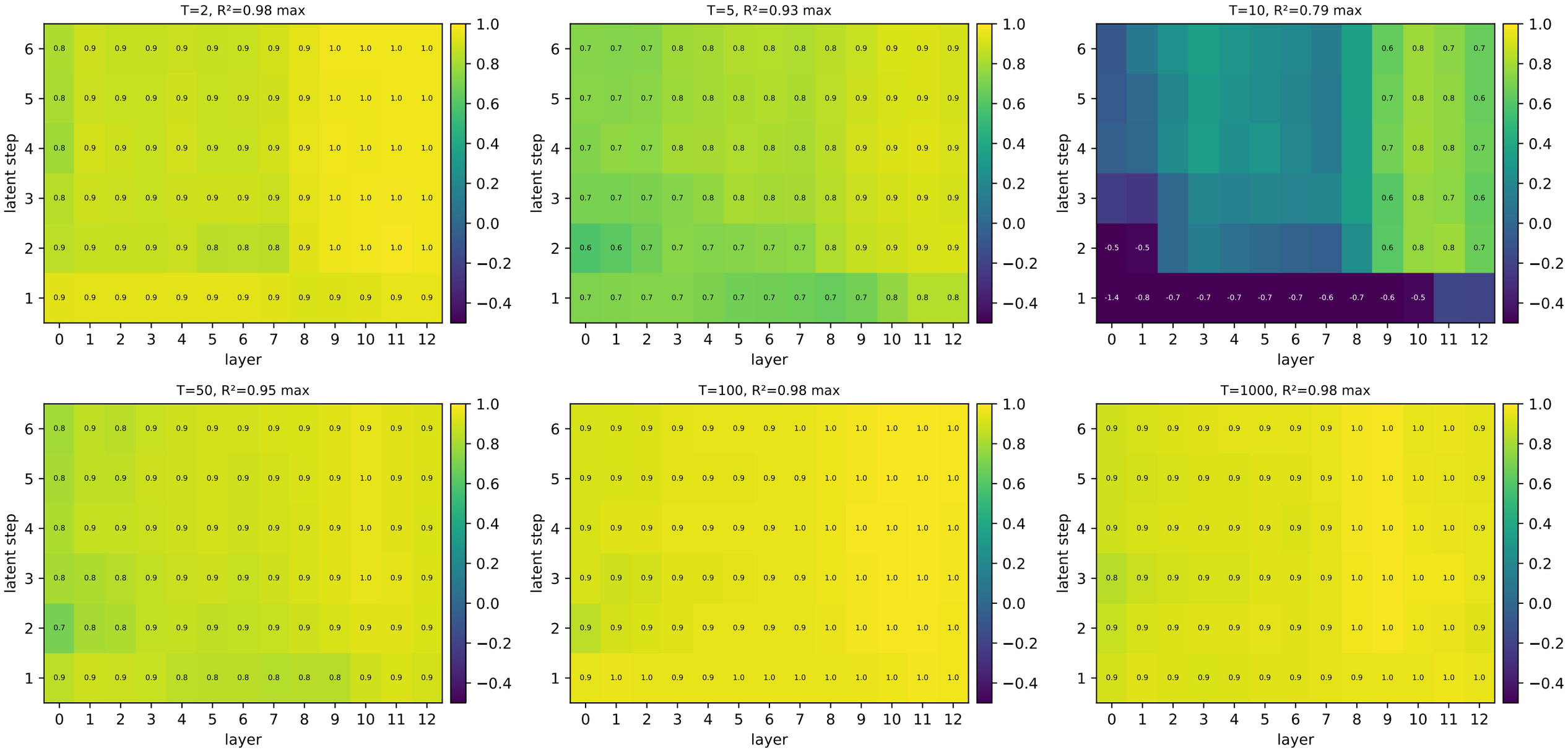
vary_b_2digit — helix R² for operand a (latent residual)

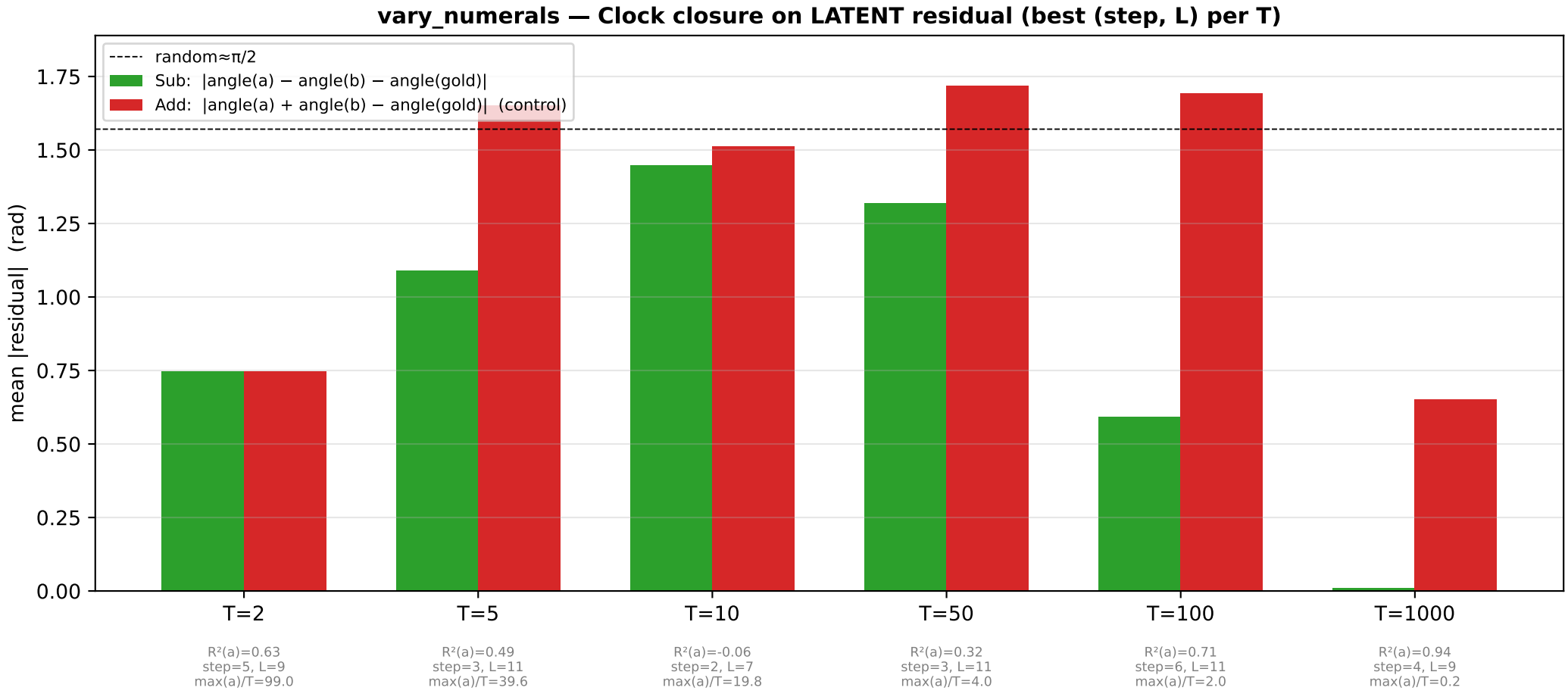


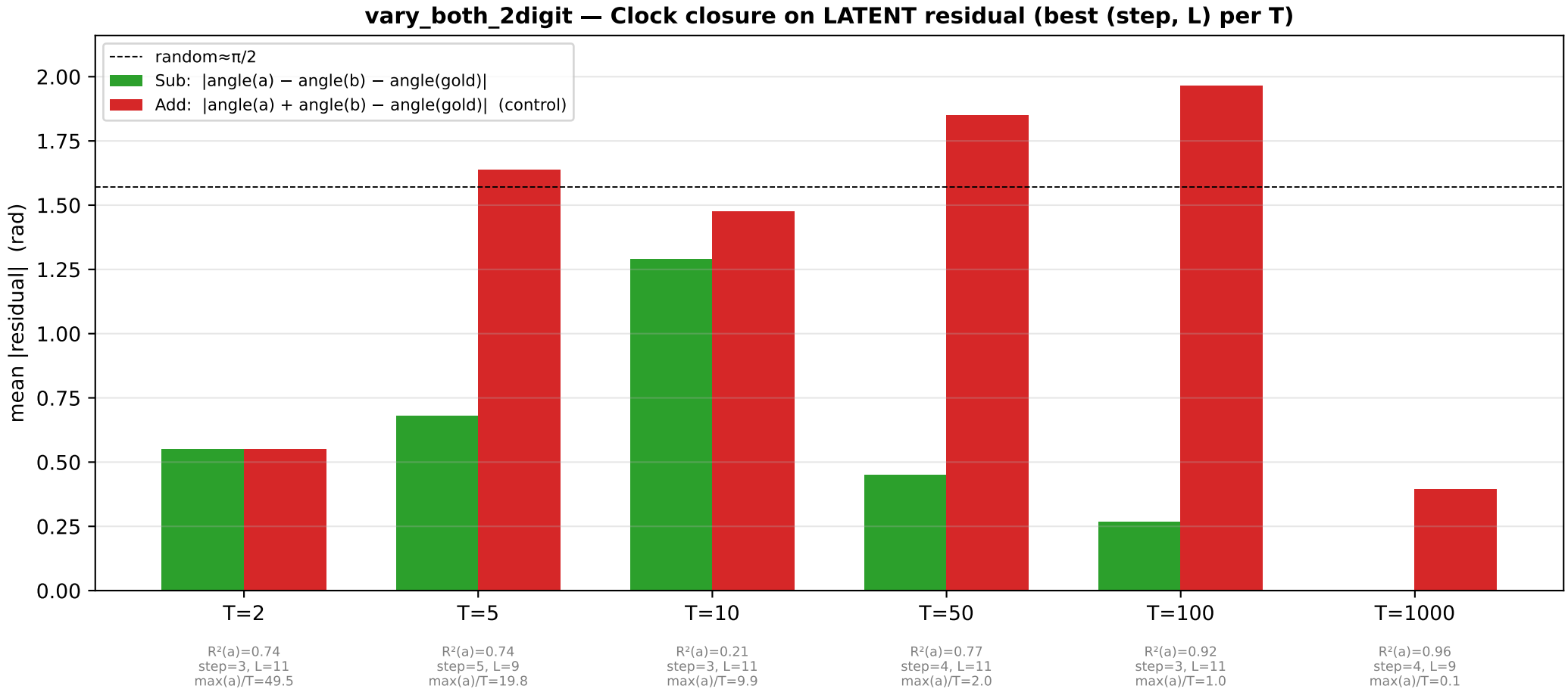
vary_b_2digit — helix R² for operand b (latent residual)

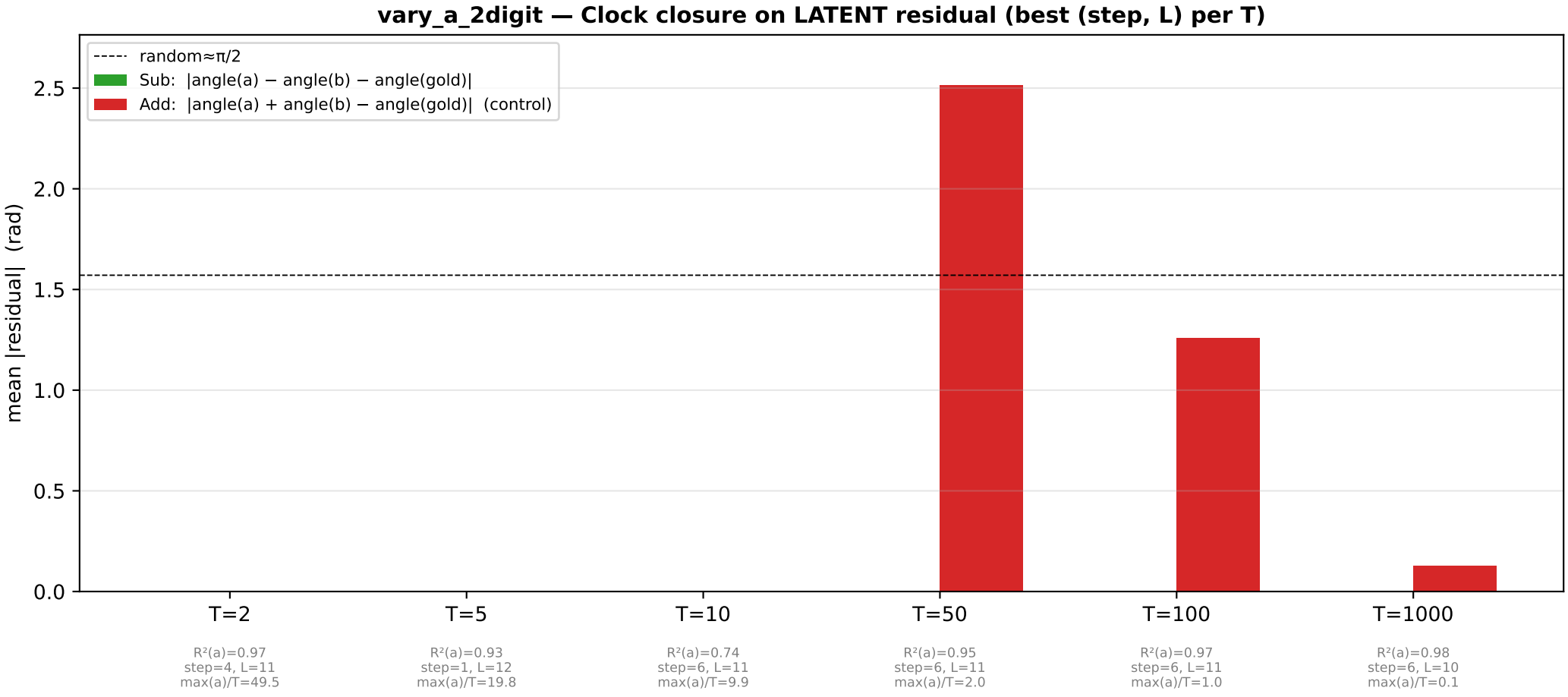


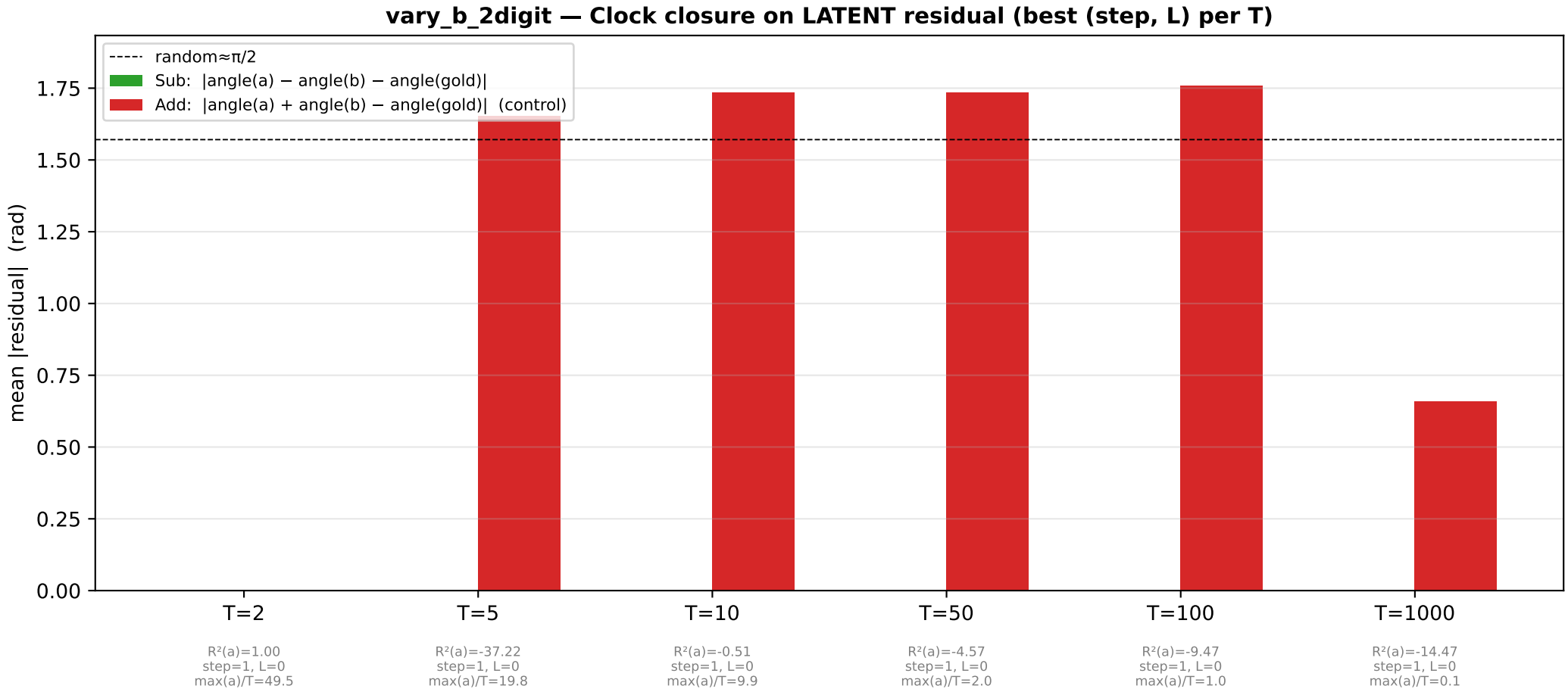
vary_b_2digit — helix R² for operand gold (latent residual)




vary_numerals — Clock closure on LATENT residual (best (step, L) per T)
random≈π/2
Sub: |angle(a) − angle(b) − angle(gold)|
Add: |angle(a) + angle(b) − angle(gold)| (control)
mean |residual| (rad)
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
T=2
T=5
T=10
T=50
T=100
T=1000
R²(a)=0.63
step=5, L=9
max(a)/T=99.0
R²(a)=0.49
step=3, L=11
max(a)/T=39.6
R²(a)=-0.06
step=2, L=7
max(a)/T=19.8
R²(a)=0.32
step=3, L=11
max(a)/T=4.0
R²(a)=0.71
step=6, L=11
max(a)/T=2.0
R²(a)=0.94
step=4, L=9
max(a)/T=0.2


vary_both_2digit — Clock closure on LATENT residual (best (step, L) per T)
random≈π/2
Sub: |angle(a) − angle(b) − angle(gold)|
Add: |angle(a) + angle(b) − angle(gold)| (control)
mean |residual| (rad)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
T=2
R²(a)=0.74
step=3, L=11
max(a)/T=49.5
T=5
R²(a)=0.74
step=5, L=9
max(a)/T=19.8
T=10
R²(a)=0.21
step=3, L=11
max(a)/T=9.9
T=50
R²(a)=0.77
step=4, L=11
max(a)/T=2.0
T=100
R²(a)=0.92
step=3, L=11
max(a)/T=1.0
T=1000
R²(a)=0.96
step=4, L=9
max(a)/T=0.1


vary_a_2digit — Clock closure on LATENT residual (best (step, L) per T)
random≈π/2
Sub: |angle(a) − angle(b) − angle(gold)|
Add: |angle(a) + angle(b) − angle(gold)| (control)
mean |residual| (rad)
2.5
2.0
1.5
1.0
0.5
0.0
T=2
R²(a)=0.97
step=4, L=11
max(a)/T=49.5
T=5
R²(a)=0.93
step=1, L=12
max(a)/T=19.8
T=10
R²(a)=0.74
step=6, L=11
max(a)/T=9.9
T=50
R²(a)=0.95
step=6, L=11
max(a)/T=2.0
T=100
R²(a)=0.97
step=6, L=11
max(a)/T=1.0
T=1000
R²(a)=0.98
step=6, L=10
max(a)/T=0.1


vary_b_2digit — Clock closure on LATENT residual (best (step, L) per T)
random≈π/2
Sub: |angle(a) − angle(b) − angle(gold)|
Add: |angle(a) + angle(b) − angle(gold)| (control)
mean |residual| (rad)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
T=2
T=5
T=10
T=50
T=100
T=1000
R²(a)=1.00
step=1, L=0
max(a)/T=49.5
R²(a)=-37.22
step=1, L=0
max(a)/T=19.8
R²(a)=-0.51
step=1, L=0
max(a)/T=9.9
R²(a)=-4.57
step=1, L=0
max(a)/T=2.0
R²(a)=-9.47
step=1, L=0
max(a)/T=1.0
R²(a)=-14.47
step=1, L=0
max(a)/T=0.1